

Complete Placement & Interview Master Roadmap

CURATED COMPLETE REVISION ROADMAP & RESOURCES SHEET

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PART I: ONLINE ASSESSMENT (OA) PREPARATION

1. Aptitude

MUST LEARN

- Percentages
- Profit & Loss
- Ratio
- Average
- Time & Work
- Time Speed Distance
- Pipes & Cisterns
- Partnership
- Probability
- Permutations & Combinations
- Simple Interest
- Compound Interest

RESOURCES

WEBSITES

<https://www.indiabix.com>

<https://prepinsta.com>

<https://testbook.com>

YOUTUBE

Rodha | MBA Pathshala | Adda247

2. Logical Reasoning

MUST LEARN

- Seating Arrangement
- Blood Relations
- Coding-Decoding
- Direction Sense
- Puzzles
- Syllogism
- Statement & Conclusion
- Calendar
- Clock

RESOURCES

WEBSITES

<https://www.indiabix.com>

<https://prepinsta.com>

<https://www.freshersnow.com>

YOUTUBE

Rodha | Testbook | TalentSprint

3. English

MUST LEARN

- Grammar
- Reading Comprehension
- Vocabulary
- Synonyms
- Antonyms
- Para Jumbles
- Sentence Correction
- Error Detection
- Cloze Test

RESOURCES

WEBSITES

<https://www.indiabix.com>

<https://learnenglish.britishcouncil.org>

<https://www.vocabulary.com>

YOUTUBE

StudyIQ English | Testbook | Adda247

4. Debugging

MUST LEARN

- Syntax Errors
- Runtime Errors
- Logical Errors
- Fix Existing Code
- Dry Run

RESOURCES

WEBSITES

<https://www.geeksforgeeks.org>

<https://www.codingnotes.io/>

<https://www.codewithharry.com/notes>

5. Coding Fundamentals

MUST LEARN

- Pointers / References / Iterators
- **Time & Space Complexity** - Big O, Big Theta, Big Omega
- Language (C++ / Java / Python) Top 10 Question
- Exception Handling
- Searching
- String Builder/StringBuffer
- Sorting (Merge, Quick, Heap, Count, Radix)
- Stack Memory vs Heap Memory
- Memory Leaks
- Garbage Collection (Java/Python)
- File Handling, Bitwise, Assignment
- Ternary Operator
- Syntax Errors, Runtime Errors
- Dynamic Memory

RESOURCES

WEBSITES

<https://www.hackerrank.com>

<https://app.codility.com/programmers/>

<https://www.geeksforgeeks.org>

<https://www.w3schools.com>

CS FUNDAMENTALS - 1. OOPS

MUST LEARN

- Four Pillars (Encapsulation, Abstraction, Inheritance, Polymorphism)
- Runtime vs Compile Time Polymorphism
- **Method Overloading vs Overriding**
- **Interface vs Abstract Class**
- Association, Aggregation, Composition
- SOLID Principles
- Constructor and Friend Function
- **Object Cloning** - Deep vs Shallow Copy
- Static vs Instance Members
- Access Modifiers
- Diamond Problem
- Dependency Injection
- Design Patterns (Singleton, Factory, Strategy)

RESOURCES

PLATFORMS

Refactoring Guru | Java Brains | Concept & Coding | CodeWithHarry

CS FUNDAMENTALS - 2. DBMS

MUST LEARN

- ACID Properties
- Normalization (1NF-BCNF)
- Denormalization
- **Primary vs Foreign Key**
- Indexing
- **Clustered vs Non-Clustered Index**
- Joins (All Types)
- Transactions
- Locks (Shared/Exclusive)
- Deadlock
- Isolation Levels
- CAP Theorem
- Sharding
- **SQL vs NoSQL**

RESOURCES

PLATFORMS

GeeksforGeeks DBMS | Gate Smashers | Neso Academy

CS FUNDAMENTALS - 3. Computer Networks

MUST LEARN

- **TCP vs UDP**
- **OSI Model** - 7 Layer
- TCP/IP Model
- Three-Way Handshake
- **HTTP vs HTTPS**
- DNS Working
- How Browser Opens Google.com
- **IPV4 VS IPV6**

RESOURCES

PLATFORMS

Neso Academy | ByteByteGo | Computer Networking Full Course FreeCodeCamp

CS FUNDAMENTALS - 4. Operating System

MUST LEARN

- **Process vs Thread**
- Context Switching
- Multithreading
- Scheduling Algorithms
- Deadlock
- Starvation
- Race Condition
- Mutex
- Semaphore
- Paging
- Segmentation
- Virtual Memory
- **Heap vs Stack**
- Memory Allocation
- Garbage Collection

RESOURCES

PLATFORMS

Neso Academy | Gate Smashers | Operating Systems Three Easy Pieces

CS FUNDAMENTALS - 5. SQL

MUST LEARN

- SELECT, WHERE
- ORDER BY
- LIMIT / TOP
- DISTINCT
- Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- GROUP BY, HAVING
- JOINS (INNER, LEFT, RIGHT)
- Subqueries
- CASE WHEN
- String Functions, Date Functions

TOP CORE QUERIES SOLUTIONS

Second Highest Salary: <https://leetcode.com/problems/second-highest-salary/>

Nth Highest Salary: <https://leetcode.com/problems/nth-highest-salary/>

Department Highest Salary: <https://leetcode.com/problems/department-highest-salary/>

Rank Scores: <https://leetcode.com/problems/rank-scores/>

Employees Earning More Than Their Managers: <https://leetcode.com/problems/employees-earning-more-than-their-managers/>

DSA Roadmap Categories

(MOST ASKED)

- Arrays
- Strings
- Two Pointers
- Binary Search
- Sorting
- Linked List (Impl + Ques)

(IMP & MUST LEARN)

- Recursion
- Stack (Impl + Ques)
- Queue & Deque (Imp + Ques)
- Sliding Window
- Heap / Priority Queue
- Greedy

(ADV. & BAR RAISER)

- Trees
- Binary Search Tree (BST) (Imp + Ques)
- Trie (Imp + Ques)
- Graphs
- Union Find (DSU)
- Dynamic Programming (DP)

RESOURCES

MENTORS & CHANNELS

Striver (Take U Forward) — <https://www.youtube.com/@takeUforward>

Pepcoding — <https://www.youtube.com/@Pepcoding>

Abdul Bari — https://www.youtube.com/@abdul_bari

Aditya Verma — <https://www.youtube.com/@AdityaVermaTheCodingTeacher>

CodeHelp by Babbar — <https://www.youtube.com/@CodeHelp>

Kunal Kushwaha — <https://www.youtube.com/@KunalKushwaha>

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Curated DSA Practice Sheets

Striver SDE Sheet (191 Problems) — <https://takeuforward.org/interviews/strivers-sde-sheet-top-coding-interview-problems/>

Striver A2Z DSA Course — <https://takeuforward.org/strivers-a2z-dsa-course/>

NeetCode 150 Roadmap — <https://neetcode.io/roadmap>

Blind 75 LeetCode Questions — <https://leetcode.com/discuss/general-discussion/460599/blind-75-leetcode-questions>

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DSA Advanced Track: DP Solutions

CORE PROBLEMS

Longest Common Subsequence (LCS) — <https://leetcode.com/problems/longest-common-subsequence/>

Longest Increasing Subsequence (LIS) — <https://leetcode.com/problems/longest-increasing-subsequence/>

Climbing Stairs — <https://leetcode.com/problems/climbing-stairs/>

House Robber — <https://leetcode.com/problems/house-robber/>

Coin Change — <https://leetcode.com/problems/coin-change/>

Unique Paths — <https://leetcode.com/problems/unique-paths/>

0/1 Knapsack (GeeksforGeeks) — <https://www.geeksforgeeks.org/problems/0-1-knapsack-problem0945/1>

Partition Equal Subset Sum — <https://leetcode.com/problems/partition-equal-subset-sum/>

DSA Advanced Track: Graphs Matrix

MUST LEARN CORE CONCEPTS

- BFS
- DFS
- Cycle Detection
- Topological Sort
- Shortest Path (Dijkstra)
- Bipartite Graph

CORE TARGET PRACTICE QUESTIONS

Number of Islands — <https://leetcode.com/problems/number-of-islands/>

Clone Graph — <https://leetcode.com/problems/clone-graph/>

Course Schedule — <https://leetcode.com/problems/course-schedule/>

Course Schedule II — <https://leetcode.com/problems/course-schedule-ii/>

Network Delay Time (Dijkstra's Algorithm) — <https://leetcode.com/problems/network-delay-time/>

Redundant Connection (Union Find / DSU) — <https://leetcode.com/problems/redundant-connection/>

Is Graph Bipartite? — <https://leetcode.com/problems/is-graph-bipartite/>

Word Ladder (BFS Shortest Path) — <https://leetcode.com/problems/word-ladder/>

LeetCode Must Do Design Problems

SYSTEM DATA BLUEPRINTING

- LRU Cache
- LFU Cache
- Min Stack
- Design Trie
- Design Hashmap
- Design Twitter
- Insert Delete GetRandom $O(1)$
- Time-Based Key-Value Store

Projects / Behavioral

CORE BEHAVIORAL PROMPTS

- Explain Your Project Architecture
- Biggest Technical Challenge
- Why This Tech Stack?
- Why Should We Hire You?
- AI Stuff
- Project Deployment & Scalability Improvements
- Security Implementations
- Performance Optimizations
- Team Conflict Situation (Story)
- Leadership Example (Story)
- Production Bug You Solved
- Future Goals

RESOURCES

PREP FRAMEWORKS

Exponent Behavioral Interviews | Amazon Leadership Principles | ByteByteGo

EXTRA TOPICS TO GO THROUGH (IF TIME LEFT)

LOW LEVEL DESIGN (LLD)

- BookMyShow
- Parking Lot
- Tic tac Toe
- URL Shortener

CORE INFRASTRUCTURE & TOOLS

- Javascript Fundamentals
- Linux
- RAG
- LLM Basics
- Git
- Redis
- Docker
- Kubernetes
- Message Queues

REFERENCE INDEX DIRECTORIES MATRIX

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Section	Best Website	Why it's the best
DSA	https://leetcode.com	Industry-standard coding practice used by most product companies
DSA Roadmap	https://neetcode.io	Curated roadmap with company-tagged questions
DSA Full Track	https://takeuforward.org/...	Complete learning + practice roadmap
SQL	https://leetcode.com/database	Best interview-focused SQL problems
SQL Interactive	https://sqlbolt.com	Interactive SQL learning
SQL Practice	https://www.hackerrank.com/sql	Company-style SQL practice
DBMS Theory	https://www.geeksforgeeks.org/dbms/	Notes + interview questions + MCQs
DBMS Board	https://www.interviewbit.com/dbms	Most asked DBMS interview questions
Operating Systems	https://www.geeksforgeeks.org/os	Theory + interview questions
OS Compilation	https://www.interviewbit.com/os	Frequently asked OS questions
Computer Networks	https://www.geeksforgeeks.org/cn	Complete networking notes
CN Compilation	https://www.interviewbit.com/cn	Interview-oriented questions
OOP	https://www.interviewbit.com/oops	Common interview questions
OOP Design	https://refactoring.guru	Design principles explained well
System Design	System Design Primer	Most recommended free resource
System Design Premium	https://bytebytego.com	Excellent explanations and visuals
Backend	https://roadmap.sh/backend	Structured backend roadmap
Backend Core	https://www.baeldung.com	Spring, Java, backend concepts

Section	Best Website	Why it's the best
Java	https://www.interviewbit.com/java	Java interview questions
Python	https://www.interviewbit.com/python	Python interview questions
JavaScript	https://javascript.info	Complete JavaScript tutorial
React	https://react.dev/learn	Official React learning
Git	https://learngitbranching.js.org	Interactive Git practice
Docker	https://docs.docker.com/docker	Official Docker tutorial
AWS	https://aws.amazon.com/aws	Official AWS learning
Linux	https://linuxjourney.com	Best beginner Linux course
REST API	https://restfulapi.net	Complete REST API concepts
HTTP	https://developer.mozilla.org/http	Best HTTP documentation
Aptitude	https://www.indiabix.com	Largest aptitude question bank
Logical Reasoning	https://www.indiabix.com/lr	Company-style reasoning questions
Verbal Ability	https://www.indiabix.com/verbal	Grammar and English practice
Quantitative Aptitude	https://www.indiabix.com/quant	Topic-wise quantitative questions
Data Interpretation	https://www.indiabix.com/di	DI questions with solutions
Behavioral / HR	https://www.interviewbit.com/hr	Common HR interview questions
Resume Questions	https://www.interviewbit.com	Project and behavioral interview guidance

Best websites for company-specific OA questions Matrix

Website	Best For
https://leetcode.com	Coding interviews, company-tagged questions
https://neetcode.io	Structured DSA preparation
https://www.geeksforgeeks.org	CS fundamentals, MCQs, interview questions
https://www.interviewbit.com	DBMS, OS, CN, OOP, HR, language interview questions
https://www.hackerrank.com	Coding, SQL, and debugging practice
https://prepinsta.com	TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini OA preparation
https://www.indiabix.com	Aptitude, reasoning, English, DI
https://www.codingninjas.com	Interview prep and OA practice
https://www.codechef.com	Competitive programming basics
https://www.codeforces.com	Advanced DSA practice
https://cses.fi/problemset/	Excellent DSA problem set
https://roadmap.sh	Backend, frontend, DevOps, and system design roadmaps